Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
PI Name:	Hulet, Randall G. Ph.D.		
Project Title:	Superfluid Phase Transition in an Ultracold Fermi Gas		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77251	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	04/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth"s gravity.		
Task Description:	Atoms have been observed to undergo Bose-Einstein condensation when the temperature is reduced to ultralow values. These atoms have integer spin and are bosons. An entirely different class of atoms, the fermions, do not exhibit this phase transition. However, fermions are predicted to form Cooper pairs, in analogy with the electronic pairing responsible for superconductivity, at very low temperatures, and to form a superfluid. We are applying the technology used to produce Bose-Einstein condensation to achieve Cooper pairing in a gas of fermions, namely lithium-6 atoms. The primary objective of this research is to detect a BCS-like phase transition in an ultracold gas of lithium-6 atoms. This type of phase transition requires that particles pair. Pairin is responsible for the phenomena of superconductivity in solids and superfluidity in liquid helium-3. It has never been observed in the gas phase, however, because densities are usually too low for pairs to form and the temperatures to high for the manifestation of quantum statistical effects. In a previously funded NASA project, we showed that lithium-6 atoms exhibit an extraordinarily large attraction for each		

Task Description:	other, which is, in fact, larger than that of any known atom. Even for lithium-6, however, the critical temperature will be ~100 nano-Kelvin. In the past few years, the techniques of laser cooling combined with evaporative cooling in a magnetic trap have been demonstrated to be effective at reaching these low temperatures in gases of bosonic atoms, where they undergo Bose-Einstein condensation. Ultracold trapped atomic gases are remarkably versatile systems for investigations of many-body quantum effects, such as Bose-Einstein condensation. Unlike condensed matter systems, many of the key parameters, such as density, interaction strength, number, and temperature are under experimental control. This ability may enable us to explore the phenomena of Cooper pairing and fermion superfluidity under conditions never before achieved. Of particular significance, is the possibility of achieving Cooper pairing in the strong coupling regime, which has not been observed experimentally, nor has it been understood theoretically.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will lead to a greater understanding of the collective behaviour of Fermi systems. This is especially important in the case of superconductors. Despite 20 years of intense effort, the exact mechanism of high-temperature superconductivity has yet to be discovered. We hope that the work that we are doing will provide insight into these important sytems.
Task Progress:	Significant progress has been made this past year. We have converted a Fermi gas of lithium-6 atoms to a diatomic molecular Bose gas. These molecular bosons can be considered to be Cooper pairs in the extreme strong coupling limit. The conversion was accomplished by sweeping a magnetic field through a resonance between pairs of free atoms and a molecular bound state. We have subsequently cooled the molecular Bose down by evaporating some of the hottest molecules out of trap which is confining them. By imaging the atoms comprising the weakly bound molecules, we have observed that the molecules have formed a Bose-Einstein condensate (BEC). A BEC is a macroscopic occupation of a single quantum mechanical energy state. The BEC is a superfluid, that is, it can flow without resistance. Our next goal is to directly image the molecules in the Bose-Einstein condensate. This will provide valuable information of the nature of the BEC. This experiment will enable us to quantitatively measure the proportion of unbound Fermi atoms as well as the deeply bound molecules forming the superfluid as a function of the magnetic field which is used to tune the interaction strength between the atoms.
Bibliography Type:	Description: (Last Updated: 02/07/2007)
Articles in Peer-reviewed Journals	Strecker, K.E.; Partridge, G.B.; Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Long-Lived Molecular Bose Gas" Physical Review Letters 91, 080406 , Aug-2003
Articles in Peer-reviewed Journals	Strecker, K.E.; Partridge, G.B.; Truscott, A.G.; Hulet, R.G. "Bright Matter Wave Solitons in Bose-Einstein Condensates" New Journal of Physics 5, 73 , Jun-2003
Articles in Peer-reviewed Journals	Prodan, I.D.; Pichler, M.; Junker, J.; Hulet, R.G.; Bohn, J.L. "Intensity Dependence of Photoassociation in a Quantum Degenerate Atomic Gas" Physical Review Letters 91, 080402 , Aug-2004
Articles in Peer-reviewed Journals	Strecker, K.E.; Partridge, G.B.; Truscott, A.G.; Hulet, R.G. "Tunable Interactions in Ultracold Bose Gases" Advances in Space Research (in press), doi:10.1016/j.asr.2003.09.058 , Jul-2004
Awards	Hulet, R.G. "Elected Member of the American Academy of Arts and Sciences" Oct-2003
Books/Book Chapters	Hulet, R.G. "Atomic Fermi Gases" in "McGraw-Hill Yearbook of Science & Technology" Ed. Staff of McGraw-Hill McGraw-Hill, 19 , Jan-2004
Dissertations and Theses	Strecker, K.E. "Tunable Interactions in Quantum Degenerate Lithium" Ph.D. Thesis, Rice University Apr-2004
Dissertations and Theses	Partridge, G.B. "An Improved System for Creating Ultracold Fermi Gases of 6Li" Masters Thesis, Rice University Sep-2003
Patents	6,657,188 issued Dec-2003 Hulet, R.G. et al. "Laser-Free Source of Free Atoms"
Presentation	Hulet, R.G. "Phase Transitions at 100 nK" Gordon Conference on Gravitational Effects in Physico-Chemical Systems Jul-2003
Presentation	Hulet, R.G. "Strongly Interacting Fermi Gas" Quantum Fluids and Solids Annual Meeting, Albuquerque Aug-2003
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Long-Lived Molecular Bose Gas" European Workshop on Bose-Einstein Condensation, San Feliu, Spain Sep-2003
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Gas of Molecular Bosons" Annual Meeting on the Physics of Quantum Electronics, Snowbird, UT Jan-2004
Presentation	Hulet, R.G. "Ultracold Atomic Gases" Meeting of the Texas Academy of Sciences, Engineering, and Medicine, San Antonio Jan-2004
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Gas of Molecular Bosons" Institute for Theoretical Atomic and Molecular Physics Workshop on Polar Molecules, Harvard University Jan-2004

Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Gas of Molecular Bosons" Physics Colloquium, University of Chicago Feb-2004
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Molecular Bose Gas" American Physical Society Annual March Meeting, Montreal Mar-2004
Presentation	Hulet, R.G. "Conversiion of an Atomic Fermi Gas to a Gas of Molecular Bosons" NASA Workshop on Fundamental Physics, Solvang, CA Apr-2004
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Gas of Molecular Bosons" Physics Colloquium, Northwestern University May-2004
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Molecular Bose-Einstein Condensate" Kavli Institute of Theoretical Physics Workshop on Quantum Gases, Santa Barbara, CA May-2004
Presentation	Hulet, R.G. "Tutorial on Bose-Einstein Condensation and Degenerate Fermi Gases" Amer. Physical Soc. Div. of Atomic, Molecular, and Optical Physics Annual Meeting, Tucson May-2004
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Long-Lived Molecular Bose Gas" Seminar, University of Innsbruck, Austria Sep-2003
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Gas of Molecular Bosons" Physics Colloquium, University of Texas, Austin Dec-2003
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Long-Lived Molecular Bose Gas" Physics Colloquium, Kansas State University, Manhattan, KS Oct-2003
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Molecular Bose Gas" Physics Colloquium, York University, Toronto Mar-2004
Significant Media Coverage	Cho, A. "Ultracold Atoms Spark a Hot Race" Science Magazine News Focus article with photograph Science 301, pg. 750, Aug. 8, 2003 , Aug-2003
Significant Media Coverage	Cho, A. "Ultracold Atoms Spark a Hot Race" Science Magazine News Focus article with photograph Science 301, pg. 750, Aug. 8, 2003 , Aug-2003